

# Project Infrastructure Specification Document

## Hyderabad Hyperscale Data Centre – Phase 1

Document No: HYD-DC-P1-IS-001 | Version 3.2 | Issued: 10 July 2026

### **HYDERABAD HYPERSCALE DATA CENTRE - PHASE 1** PROJECT INFRASTRUCTURE SPECIFICATION DOCUMENT

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|-----------------------|---------------------------------------------------------------|
| <b>Document No.</b>   | HYD-DC-P1-IS-001                                              |
| <b>Version</b>        | 3.2 (Issued for Construction)                                 |
| <b>Date of Issue</b>  | 10 July 2026                                                  |
| <b>Prepared By</b>    | Infrastructure Design Authority, Project Engineering Cell     |
| <b>Classification</b> | Confidential – Internal & Authorized Vendor Distribution Only |
| <b>Approved By</b>    | Head of Infrastructure Delivery                               |

## 1. Project Overview

| Parameter               | Detail                                                                   |
|-------------------------|--------------------------------------------------------------------------|
| Project Name            | Hyderabad Hyperscale Data Centre – Phase 1                               |
| Site Location           | Adibatla-Maheshwaram Industrial Corridor, Rangareddy District, Telangana |
| IT Load Capacity        | 45 MW (critical IT load, Phase 1 only)                                   |
| Target Uptime Tier      | Tier III (Concurrently Maintainable)                                     |
| Facility Type           | Wholesale / Hyperscale Colocation                                        |
| Total Built-Up Area     | Approx. 3,85,000 sq. ft. (across 4 data halls + support blocks)          |
| Number of Data Halls    | 4 (Phase 1), 11.25 MW average per hall                                   |
| Design PUE (Annualized) | ≤ 1.35                                                                   |
| Seismic Zone            | Zone II (per IS 1893)                                                    |
| Site Area               | 22 acres (leasehold, industrial land use classification)                 |

### 1.1 Scope of this Document

This specification defines the technical baseline for power architecture, mechanical cooling, structural rack layout, structured cabling, physical security, and fire suppression systems for Phase 1 of the facility. It is issued for construction (IFC) and supersedes all prior revisions. Any deviation requires a formal Engineering Change Note (ECN) approved by the Infrastructure Design Authority.

## 2. Site Details

- **Grid Connectivity:** Dual independent 132 kV feeders from TSSPDCL (Telangana State Southern Power Distribution Company) substations at Adibatla and Maheshwaram, terminating at an on-site 132/33 kV owner-operated substation.
- **Fiber Connectivity:** Dual-entry, physically diverse duct routes into a split Meet-Me Room (MMR-A / MMR-B), with a minimum of four carrier-neutral long-haul providers contracted at handover.
- **Water Source:** Borewell + municipal tanker backup, feeding a 2-day reserve raw water storage tank sized for cooling tower make-up under N+1 chiller operation.
- **Flood Risk:** Site finished floor level (FFL) set at +0.9 m above the 100-year flood contour per local hydrological survey; all critical electrical plant rooms located at or above ground floor level (no critical load in basement).
- **Access Roads:** Dual independent access roads from the Outer Ring Road (ORR) service lane, sized for 40T DG/transformer delivery vehicles.

## 3. Power Architecture - 2N Redundancy

### 3.1 Design Philosophy

The facility is designed on a **2N distribution topology** from the UPS output board through to the rack, meaning two fully independent, simultaneously active power trains (Train A and Train B) are provided end-to-end. This exceeds the minimum Tier III requirement (which mandates only concurrent maintainability, N+1) and is adopted here at customer request to support future Tier IV certification pursuit in a later phase.

### 3.2 Utility & Standby Generation

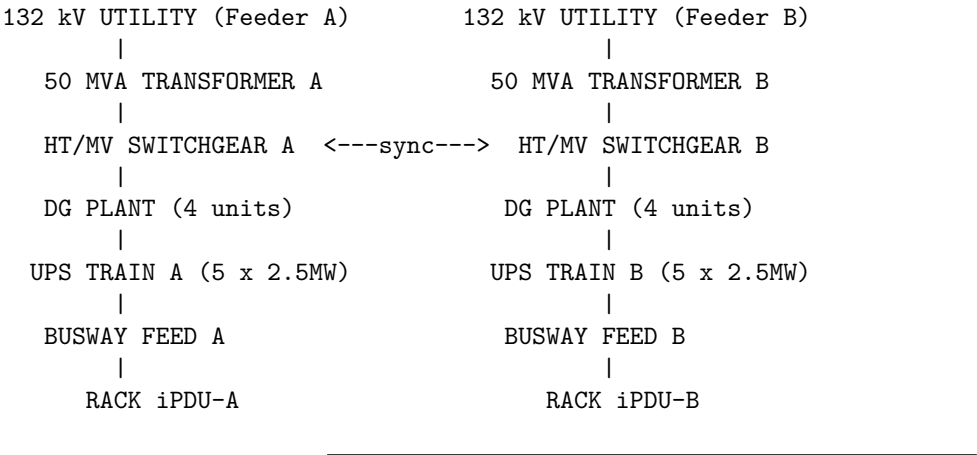
| Parameter                 | Specification                                                                         |
|---------------------------|---------------------------------------------------------------------------------------|
| Utility Intake            | 2 x 132 kV independent feeders, N+1 at substation level                               |
| Step-down Transformers    | 2 x 50 MVA, Dyn11 vector group, ONAN/ONAF cooled                                      |
| Diesel Generators         | 8 x 3 MW (3,750 kVA prime rating), N+1 configuration per data hall block              |
| Fuel Storage              | Bulk day tank + underground storage, 12-hour full-load runtime at site, PESO-licensed |
| Generator Step-Load Class | ISO 8528-5 Class G3 minimum                                                           |
| Paralleling Architecture  | Closed-transition, synchronized bus with automated load-sharing PLC                   |

### 3.3 UPS & Distribution

| Parameter                    | Specification                                                    |
|------------------------------|------------------------------------------------------------------|
| UPS Topology                 | Static, double-conversion online                                 |
| UPS Capacity (Phase 1 total) | 10 x 2.5 MW modules, deployed as 2N across 4 data halls          |
| Battery Type                 | Lithium-Iron-Phosphate (LFP), 8-minute ride-through at full load |
| UPS Output Distribution      | Dual A/B overhead busway (Starline-type track busway) to RPPs    |

| Parameter       | Specification                                                                  |
|-----------------|--------------------------------------------------------------------------------|
| Rack Power Feed | Dual-corded IT load fed directly A+B;<br>single-corded load via rack-level STS |
| Earthing System | TN-S throughout critical distribution, per IS 3043                             |

3.4 Simplified Power Chain



4. Cooling System - N+1 Configuration

| Parameter                                   | Specification                                                                   |
|---------------------------------------------|---------------------------------------------------------------------------------|
| Cooling Architecture                        | N+1 chilled water plant with dedicated CRAH units per hall                      |
| Chiller Type                                | Water-cooled centrifugal, VSD-equipped                                          |
| Chiller Configuration                       | 6 x 2,500 TR chillers (5 duty + 1 standby, N+1)                                 |
| Cooling Towers                              | 8-cell induced-draft, VFD fan control                                           |
| CRAH Units                                  | 96 units across 4 halls (N+1 per hall zone), hot-aisle containment              |
| Design Supply Air Temp                      | 22 degrees C +/- 1 degree C (ASHRAE A1 recommended envelope)                    |
| Design Rack Density (avg)                   | 8 kW/rack (air-cooled zones)                                                    |
| Design Rack Density (high-density GPU zone) | Up to 65 kW/rack, Direct-to-Chip liquid cooling                                 |
| Liquid Cooling CDU                          | 12 units, secondary loop isolated from facility primary loop                    |
| Free Cooling                                | Waterside economizer, estimated 900+ hours/year in Hyderabad climate            |
| Leak Detection                              | Continuous rope-type sensors under all D2C manifolds with auto-isolation valves |

5. Rack Layout

| Parameter                               | Specification                                                           |
|-----------------------------------------|-------------------------------------------------------------------------|
| Standard Rack Footprint                 | 19-inch EIA, 42U, 800mm x 1200mm                                        |
| High-Density Zone Racks                 | OCP ORv3, 21-inch, 52U equivalent                                       |
| Aisle Configuration                     | Hot-aisle containment (HAC), 1.2m cold aisle / 0.9m contained hot aisle |
| Raised Floor Height                     | 900mm (standard halls)                                                  |
| Floor Loading – Data Hall               | 12.5 kN/sq.m (design live load)                                         |
| Floor Loading – Battery/Immersion Zones | 27 kN/sq.m                                                              |
| Racks per Hall (avg)                    | 620 standard-density equivalent racks                                   |
| Rack Numbering Convention               | Hall-Row-Position (e.g., DH1-R04-018)                                   |

## 6. Structured Cabling Standards

- **Governing Standard:** TIA-942 / ANSI-BICSI-002, with labeling per ANSI/TIA-606-C.
- **Backbone:** Singlemode OS2 fiber between MDA and HDA; MPO/MTP-24 trunk assemblies for high-density spine-leaf interconnects.
- **Horizontal:** OM4 multimode fiber and Category 6A copper for out-of-band management, terminated at ZDA where applicable.
- **Topology:** Spine-leaf (Clos) fabric; every leaf switch connects to every spine switch within a hall pod.
- **AI Fabric:** Dedicated InfiniBand/RoCEv2 cabling for GPU-dense zones, routed on separate basket tray with bend-radius-compliant routing (minimum 10x cable OD).
- **Pathway Separation:** Power and data cable trays maintained at minimum 300mm parallel separation; crossings at 90 degrees only.
- **Cable Tray Fill:** Not to exceed 50% fill ratio at handover to allow for growth.

## 7. Security Requirements

| Layer              | Controls                                                                                         |
|--------------------|--------------------------------------------------------------------------------------------------|
| Perimeter          | Anti-climb fencing (2.4m), K12-rated crash bollards at all vehicle entries, ANPR + UVSS at gate  |
| Facility Access    | Biometric (iris + fingerprint) mantrap entry, 100% CCTV coverage with AI tailgating analytics    |
| White Space        | Multi-factor badge + biometric access, electronic rack-level locking with individual audit trail |
| Visitor Management | Pre-authorized visitor logging, escort-mandatory policy in all white space                       |
| Monitoring         | 24x7 manned SOC/NOC, minimum 90-day CCTV retention                                               |

## 8. Fire Suppression Requirements

| System                            | Specification                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------|
| Early Warning                     | VESDA aspirating smoke detection in all white space and below-raised-floor void                  |
| Primary Suppression (White Space) | Novec 1230 clean agent, engineered per room integrity door-fan test                              |
| Secondary Suppression             | Double-interlock pre-action sprinklers                                                           |
| Fuel Farm / MMR Separation        | 4-hour fire-rated compartment walls                                                              |
| General Compartmentation          | 2-hour fire-rated walls between major functional zones                                           |
| Penetration Sealing               | Intumescent fire-stop at all cable/conduit wall penetrations, verified by third-party inspection |

9. Compliance References

NBC 2016 (Part 4 Fire & Life Safety) | CEA Safety Regulations | IS 3043 (Earthing) | IS 1893 (Seismic) | IS/IEC 62305 (Lightning Protection) | TIA-942 | ANSI/BICSI-002 | ASHRAE TC 9.9 | Uptime Institute Tier Standard: Topology | NFPA 855 (referenced for Li-ion battery room design)

End of Document - HYD-DC-P1-IS-001, Version 3.2